A RATIONAL COUNTEREXAMPLE TO THE RECIPROCAL-LOGARITHMIC ASSERTION IN ERDŐS PROBLEM 143

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ABSTRACT. We construct a countably infinite set $A \subset \mathbb{Q} \cap [2, \infty)$ with $|ka - b| \geq 1$ for all distinct $a, b \in A$ and all positive integers k, but $\sum_{a \in A} 1/(a \log a) = \infty$. This gives a negative answer to the convergence question in Erdős problem 143. The proof is a finite extension argument: each step preserves all earlier points and increases the sum by at least a fixed positive constant. New points are added in blocks, each a rational dilation of a finite primitive set of integers. Two primes in each dilation factor serve different roles: one controls separation from later blocks, while the other makes the displacement from integers as small as needed. The required primitive sets are selected from periodic sets by an elementary Euler-product argument. Every constructed point lies within $m^{-12}/2$ to the right of an integer m. A companion Lean 4 project proves the existence theorem and the periodic-selection lemma.

1. INTRODUCTION

Write $\mathbb{N} = \{1, 2, \ldots\}$, and let log denote the natural logarithm. For $x > 1$ and a finite or countable set $B \subset (1, \infty)$, write

$$\mathrm{w}(x) = \frac{1}{x \log x}, \qquad W(B) = \sum_{x \in B} \mathrm{w}(x) \in [0, \infty].$$

The sum is the supremum of its finite subsums. We call $A \subset (1, \infty)$ *dilation-separated* if

$$|ka - b| \geq 1 \qquad (a, b \in A,\ a \neq b,\ k \in \mathbb{N}). \tag{1.1}$$

Both orders of each pair are included. In particular, taking $k = 1$ shows that distinct points are at least one apart. Such a set has only finitely many points in every bounded interval and is at most countable.

For a set of integers greater than one, (1.1) is equivalent to *primitivity*: no element divides another distinct element. Erdős proved that every primitive set of integers has finite weight W [2]. In problem II of [3, p. 692], he asked whether the analogous separation condition for real numbers implies both

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convergence of $W(A)$ and vanishing logarithmic density. These are the two questions in problem 143 of [1]. Koukoulopoulos, Lamzouri, and Lichtman [4, Theorem 1] proved that every dilation-separated set satisfies

$$\sum_{a\in A\cap[1,x]} \frac{1}{a} = o(\log x). \tag{1.2}$$

We give a counterexample to the convergence assertion. Thus the two questions have different answers, even when all points are rational and extremely close to integers.

Theorem 1.1. *There exists a countably infinite set $A \subset \mathbb{Q} \cap [2,\infty)$ with no finite accumulation point such that* (1.1) *holds and*

$$W(A) = \infty. \tag{1.3}$$

Every $a \in A$ can be written as

$$a = m + \eta, \qquad m \in \mathbb{N}, \quad 0 < \eta \le \frac{1}{2} m^{-12}. \tag{1.4}$$

More precisely, the construction produces one nested sequence of finite dilation-separated sets $F_1 \subset F_2 \subset \cdots$ whose union is A and for which

$$W(F_R) \ge \frac{R}{576} \qquad (R \in \mathbb{N}). \tag{1.5}$$

The construction must preserve separation from every integer multiple of every earlier point while adding an infinite total weight. Separation within one new finite set is easy. If P is primitive and $c > 1$, then $|ckn - cn'| \ge c$ for distinct $n, n' \in P$ and $k \in \mathbb{N}$. The difficulty is making many such sets compatible with each other.

We use multipliers of the form

$$D\frac{pq}{pq-1} = D + \frac{D}{pq-1},$$

where D is a positive integer and p, q are primes. The prime q is chosen last, so it can make every displacement in the new finite set as small as required. The prime p controls later exclusions: on a suitable new integer grid, every location not divisible by p stays more than 1 away from every integer multiple of every point in this block. Definition 2.2 records the divisibility conditions that make this possible. Section 2 proves the finite extension in Proposition 2.3; iterating it gives Theorem 1.1.

The only analytic input to this construction is that a periodic set of integers of density greater than $2/3$ contains, above any prescribed threshold, a finite primitive subset of weight greater than $1/6$. We state this as Lemma 2.1 before the construction and prove it in Section 3. The constants and the exponent 12 are convenient choices; no optimality is claimed.

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2. The finite extension argument

A set $E \subset \mathbb{N}$ is *periodic* if membership depends only on the residue modulo some positive integer L. Its density $\mathrm{d}(E)$ is the proportion of allowed residue classes modulo L; this proportion is independent of the chosen period.

Lemma 2.1 (Periodic selection)**.** *Let $E \subset \mathbb{N}$ be periodic, with $\mathrm{d}(E) > 2/3$. For every integer $N \geq 2$, there is a finite primitive set $P \subset E \cap [N, \infty)$ such that $W(P) > 1/6$.*

The proof of this lemma is given in Section 3. We now show how it gives the finite extension needed for the theorem.

2.1. Admissible finite configurations.

A block has an integer $D_i \geq 48$, a finite nonempty primitive set $P_i \subset \mathbb{N} \setminus \{1\}$, and two primes p_i, q_i. Its points are

$$A_i = \{a_{i,n} : n \in P_i\}, \qquad a_{i,n} = \frac{D_i n p_i q_i}{p_i q_i - 1} = D_i n + \frac{D_i n}{p_i q_i - 1}. \tag{2.1}$$

Thus a block is a common rational dilation of $D_i P_i$, with a positive displacement that can be made arbitrarily small by increasing the auxiliary prime q_i. We call p_i the *private prime* of the block: it does not divide $p_i q_i - 1$, and the following condition prevents it from dividing $q_j D_j n$ in any block, including its own.

Definition 2.2. For a finite ordered list of blocks, write $F = \bigcup_{i=1}^{t} A_i$. We call the list *admissible* if the following conditions hold.

(i) F is dilation-separated, and every point of A_i exceeds every point of A_j by more than 1 whenever $j < i$.

(ii) The primes p_i are distinct and greater than 2, and

$$p_j \nmid q_i D_i n \qquad (1 \leq i, j \leq t,\ n \in P_i). \tag{2.2}$$

(iii) For every i and $n \in P_i$,

$$\frac{a_{i,n}}{p_i} > 4, \qquad 0 < a_{i,n} - D_i n \leq \frac{1}{2}(D_i n)^{-12}. \tag{2.3}$$

(iv) The private primes obey the strict bound

$$\sum_{i=1}^{t} \frac{1}{p_i} < \frac{1}{100}. \tag{2.4}$$

The empty list is admissible. Put $M(F) = \max F$ when $F \neq \varnothing$, and $M(\varnothing) = 0$.

The conditions serve two purposes. Condition (i) records the separation already achieved. Conditions (ii)–(iv) make further extensions possible: (ii) allows a common multiple of all the integers $q_i D_i n$ that is divisible by none of the private primes; the first bound in (iii) then gives a separation margin, while (iv) keeps the excluded proportion of integers small. The displacement bound in (iii) records the additional conclusion of Theorem 1.1. We retain all these arithmetic data when extending a configuration.

Proposition 2.3 (Finite extension)**.** *Every admissible finite configuration with point set F can be extended by appending finitely many blocks to an admissible configuration with point set F' such that*

$$W(F' \setminus F) \geq \frac{1}{576}.$$

All preceding block data, and hence every point of F, remain unchanged.

One application of this proposition will add several blocks with a common integer D; we call this group of insertions a *round*. The gain from a single block depends on D, whereas the gain from a complete round has the fixed lower bound above. We first deduce the theorem by iterating complete rounds, then prove that each round can be carried out.

Proof of Theorem 1.1. Start with the empty configuration and successively choose an extension given by Proposition 2.3. If F_R is the point set after R extensions, then these choices give one nested sequence

$$F_0 = \varnothing \subset F_1 \subset F_2 \subset \cdots, \qquad W(F_R) \geq \frac{R}{576}.$$

Set $A = \bigcup_{R\geq 1} F_R$. Every point is rational and greater than 2, by (2.1). Given distinct $a, b \in A$, both belong to some F_R, which satisfies $|ka - b| \geq 1$ for every $k \in \mathbb{N}$. Thus A is dilation-separated. The same R works for every multiplier; no limiting argument in k is involved.

The set A is countable, and its separation for $k = 1$ implies local finiteness. The displacement bound (1.4) follows from (2.3), since each inserted point is retained with its original data. Finally,

$$W(A) \geq W(F_R) \geq R/576 \qquad (R \geq 1),$$

so $W(A) = \infty$ and A is infinite. The sets F_R also satisfy (1.5). □

2.2. **Choosing the next integer scale.** To prove Proposition 2.3, we seek new points near integer locations $D\ell$, where $\ell \in \mathbb{N}$. We call $D\mathbb{N}$ the grid and ℓ the label of a location. The following lemma chooses D so that the preceding blocks exclude only labels divisible by their private primes. The preceding values D_i and all old points stay fixed.

Lemma 2.4. *For every admissible configuration with point set F_0, there is an integer $D \geq 48$ such that*

$$q_i D_i n \mid D \quad (n \in P_i), \qquad p_i \nmid D \quad (1 \leq i \leq t). \tag{2.5}$$

On this grid, if $\ell \in \mathbb{N}$ and $p_i \nmid \ell$, then

$$|k a_{i,n} - D\ell| > 4 \qquad (n \in P_i,\ k \in \mathbb{Z}). \tag{2.6}$$

If $p_i \mid \ell$, then $D\ell$ is an integer multiple of every point in A_i.

Proof. Take the product of the finitely many integers $q_i D_i n$ and multiply it by a sufficiently large power of 2. The resulting integer D is at least 48 and is divisible by every $q_i D_i n$. By (2.2), and because all p_i are odd, none of the p_i divides D. For the empty configuration take $D = 48$.

Fix $a = a_{i,n}$, and write $v = D/(q_i D_i n)$ for the corresponding integer quotient. Then

$$|ka - D\ell| = \frac{a}{p_i} \left| k p_i - v(p_i q_i - 1)\ell \right|. \tag{2.7}$$

Since $p_i \nmid v$ and $p_i q_i - 1 \equiv -1 \pmod{p_i}$, the integer inside the absolute value is nonzero when $p_i \nmid \ell$. Its absolute value is therefore at least 1, and (2.3) gives (2.6). When $p_i \mid \ell$, the ratio $D\ell/a = v(p_i q_i - 1)\ell/p_i$ is an integer. □

Consequently, the labels ℓ avoiding the multiples of all preceding private primes have density at least $1 - \sum_i p_i^{-1} > 99/100$. For these labels, $D\ell$ stays more than 4 away from every integer multiple of an old point. A separate estimate controls the blocks that will be inserted before this grid is changed again.

2.3. **The density excluded by one block.** Once D has been fixed, we must also avoid multiples of the new blocks inserted on this grid. All multiples of a block are multiples of the single rational number $Dpq/(pq-1)$. The following set contains all residue classes in which a distance of at most 1 can occur. For $D \geq 1$ and primes p, q with $pq \geq 2D$, let

$$\mathcal{B}_{p,q}(D) = \left\{ z \bmod pq : z \in \mathbb{Z}, \quad |z| \leq \left\lfloor \frac{pq-1}{D} \right\rfloor \right\}. \tag{2.8}$$

The bound below is independent of the number of points in the block.

Lemma 2.5. *If $\ell \in \mathbb{N}$ and $\ell \bmod pq \notin \mathcal{B}_{p,q}(D)$, then*

$$\left| \frac{Dpq}{pq-1} h - D\ell \right| > 1 \qquad (h \in \mathbb{Z}).$$

Moreover,

$$\frac{|\mathcal{B}_{p,q}(D)|}{pq} \leq \frac{3}{D}. \tag{2.9}$$

Proof. If the displayed absolute value is at most 1, then the integer $z = pqh - (pq-1)\ell$ satisfies $D|z| \leq pq - 1$. But $z \equiv \ell \pmod{pq}$, so the residue

of ℓ belongs to $\mathcal{B}_{p,q}(D)$. For the density bound, reduction modulo pq cannot increase cardinality, and hence

$$\frac{|\mathcal{B}_{p,q}(D)|}{pq} \le \frac{2\lfloor (pq-1)/D \rfloor + 1}{pq} \le \frac{2}{D} + \frac{1}{pq} \le \frac{3}{D}. \qquad \square$$

For a block with $D_i = D$, the lemma applies to every $ka_{i,n}$ by taking $h = kn$. The number $Dp_iq_i/(p_iq_i - 1)$ is used only to count these multiples; it is not inserted into the configuration.

2.4. **Proof of the finite extension.**

Proof of Proposition 2.3. Let the initial configuration consist of t blocks, with point set F_0. Choose D by Lemma 2.4. We keep D fixed while adjoining

$$J = \lfloor D/12 \rfloor$$

blocks. By Lemma 2.5, their residue exclusions have total density at most $3J/D \le 1/4$. We will also show that each new block has weight greater than $1/(24D)$, so a full round compensates for the loss of weight caused by moving to the scale D. After s insertions, where $0 \le s \le J$, we require the following local induction conditions:

(1) the current configuration is admissible and retains all preceding block data;
(2) $D_j = D$ and $p_jq_j \ge 2D$ for $t < j \le t + s$;
(3) $p_j \nmid D$ for every private prime currently present;
(4) its point set F satisfies $W(F) \ge W(F_0) + s/(24D)$.

These hold for $s = 0$. Suppose they hold for $s < J$, and let $i = t + s + 1$ be the next block index.

The private prime and the available integers. Choose a prime p_i greater than D, all preceding private primes, all the preceding integers q_jD_jn, and

$$\left(\frac{1}{100} - \sum_{j<i} \frac{1}{p_j} \right)^{-1}.$$

Admissibility makes the quantity in parentheses positive, and all the bounds are finite. This choice preserves the strict reciprocal bound and ensures that p_i divides neither D nor any preceding q_jD_jn.

Define the periodic set

$$E = \left\{ \ell \in \mathbb{N} : \begin{array}{ll} p_j \nmid \ell & (1 \le j \le i), \\ \ell \bmod p_jq_j \notin \mathcal{B}_{p_j,q_j}(D) & (t < j < i) \end{array} \right\}. \tag{2.10}$$

The first row excludes multiples of all preceding private primes and of the newly chosen p_i. It handles the initial blocks; the second row handles those already inserted on the same grid. Take the product of the displayed moduli

as a common period. For each modulus, its residue classes have equally many lifts to this period, so the union bound and Lemma 2.5 give

$$\mathrm{d}(E) \geq 1 - \sum_{j=1}^{i} \frac{1}{p_j} - \sum_{j=t+1}^{i-1} \frac{|\mathcal{B}_{p_j,q_j}(D)|}{p_j q_j} > \frac{99}{100} - \frac{3s}{D} \geq \frac{37}{50} > \frac{2}{3}. \tag{2.11}$$

No coprimality of the moduli is needed, and each private prime is counted only once. Lemmas 2.4 and 2.5, applied respectively to the initial blocks and those inserted in this round, give

$$|ka - D\ell| > 1 \qquad (a \in F,\ \ell \in E,\ k \in \mathbb{Z}). \tag{2.12}$$

The set E, including its period, is now fixed. By Lemma 2.1, choose a finite primitive $P_i \subset E$ such that

$$\begin{gathered} W(P_i) > \frac{1}{6}, \qquad \min P_i \geq 2D, \\ D \min P_i > \max\{4p_i, M(F) + 1\}. \end{gathered} \tag{2.13}$$

The bound $n \geq 2D$ will give the weight comparison at the end of the step. The other two bounds ensure $a_{i,n}/p_i > 4$ and place the new block beyond every old point. All can be imposed through the threshold in Lemma 2.1, since F, D, p_i are already fixed. The positive weight ensures that P_i is nonempty.

The auxiliary prime and the displacement. Put $U = D \max P_i$. For each $a \in F$, write $a = A_a/B_a$ in reduced form, with $B_a > 0$. Choose a prime q_i, different from all $p_1, \ldots, p_i$, so large that

$$\begin{gathered} p_i q_i \geq 2D, \qquad p_i q_i - 1 \geq 2U^{13}, \\ \frac{U}{p_i q_i - 1} < \frac{1}{B_a} \quad (a \in F). \end{gathered} \tag{2.14}$$

These are finitely many lower bounds, so such a prime exists. Set $D_i = D$ and define the block by (2.1). For $m = Dn$ with $n \in P_i$, its displacement is

$$a_{i,n} - m = \frac{m}{p_i q_i - 1} \leq \frac{m}{2U^{13}} \leq \frac{1}{2} m^{-12}. \tag{2.15}$$

The auxiliary prime is chosen after the finite set P_i precisely so that it can meet all these bounds without changing E or P_i.

Separation. Fix an old point $a = A_a/B_a$, a new point $a_{i,n}$, and $k \in \mathbb{Z}$. With $m = Dn$, equation (2.12) and the fact that $ka - m \in B_a^{-1}\mathbb{Z}$ give the alternatives

$$ka - m < -1 \qquad \text{or} \qquad ka - m \geq 1 + \frac{1}{B_a}.$$

The displacement in (2.15) is positive and, by (2.14), strictly smaller than $1/B_a$. Subtracting it preserves a distance greater than 1 in both cases. Thus $|ka - a_{i,n}| > 1$ for every integer k, with no bound on its size.

In the opposite direction, (2.13) gives $a_{i,n} > M(F) + 1$, so $ka_{i,n} - a > 1$ for every $k \in \mathbb{N}$.

For distinct $n, n' \in P_i$ and $k \in \mathbb{N}$, primitivity gives $kn \neq n'$. Hence, within the new block,

$$|ka_{i,n} - a_{i,n'}| = \frac{Dp_iq_i}{p_iq_i - 1}|kn - n'| > D > 1.$$

This proves the required separation and ordering of the blocks.

Preserving the arithmetic conditions. For (2.2), the new p_i exceeds every old q_jD_jn, while the preceding private primes already avoid these integers. For a new integer q_iDn, every p_j with $j \leq i$ avoids each of its three factors: it avoids q_i by the choice of the auxiliary prime, D by induction and $p_i > D$, and n by $P_i \subset E$. Since p_j is prime, $p_j \nmid q_iDn$. Finally, $a_{i,n} > Dn > 4p_i$, and (2.15) verifies the remaining part of (2.3). All admissibility conditions and the first three local induction conditions are preserved.

Weight and completion of the round. Since $1 < p_iq_i/(p_iq_i-1) < 2$ and $n \geq 2D$, we have $a_{i,n} < 2Dn \leq n^2$, and hence $\log a_{i,n} \leq 2\log n$. Consequently

$$\mathrm{w}(a_{i,n}) \geq \frac{\mathrm{w}(n)}{4D}, \qquad W(A_i) > \frac{1}{24D}.$$

The map $n \mapsto a_{i,n}$ is injective, so summing the first inequality over P_i and using (2.13) gives the second. Disjointness from F gives the fourth local induction condition for $F \cup A_i$. This completes the insertion step.

Finite induction now supplies all J blocks. Let F' be the resulting point set. Since $D \geq 48$,

$$J \geq D/12 - 1 \geq D/24, \qquad W(F' \setminus F_0) \geq \frac{J}{24D} \geq \frac{1}{576}.$$

The construction has changed none of the preceding block data. Its final admissibility allows Lemma 2.4 to be applied again at the start of the next round, using every integer q_iD_in then present. The new grid need not equal D; it affects only future blocks. □

3. Primitive sets in periodic sets

We prove Lemma 2.1. For $n \in \mathbb{N}$, let $\Omega(n)$ count prime factors with multiplicity, with $\Omega(1) = 0$. Integers with the same positive value of Ω form a primitive set: if n properly divides n', then $\Omega(n') > \Omega(n)$. We seek arbitrarily large levels whose weight in E exceeds $1/6$. For $0 < u < 1$, we average the weights of the levels $\Omega(n) = r$, $r \geq 0$, using the coefficients $(1-u)u^r$, whose sum is 1, and then let u tend to 1 from below.

For a fixed modulus $L \geq 1$, residue $b \pmod L$ and $0 < u < 1$, set

$$A_{L,b}(u) = \sum_{\substack{n \geq 2 \\ n \equiv b \pmod L}} \frac{u^{\Omega(n)}}{n \log n}.$$

We first justify convergence and prove

$$\liminf_{u \uparrow 1} (1-u) A_{L,b}(u) \geq \frac{1}{4L}. \tag{3.1}$$

Summing this estimate over the allowed residue classes of E will give a lower bound greater than $1/6$ for the normalized average. We then extract a finite set from a sufficiently high level. All limits are taken with L fixed.

3.1. **From logarithmic weights to Euler products.** Dirichlet characters will impose the congruence condition for coprime residues. To obtain the weight $1/(n\log n)$ from an Euler product, we use the identity

$$\frac{1}{n\log n}=\int_0^\infty n^{-1-t}\,dt \qquad (n\ge 2).$$

For a Dirichlet character χ modulo L and $t>0$, define

$$G_\chi(u,t)=\prod_{p\text{ prime}}(1-u\chi(p)p^{-1-t})^{-1}=\sum_{n\ge1}\frac{u^{\Omega(n)}\chi(n)}{n^{1+t}}.$$

These products and series converge absolutely. Expanding the logarithms of their Euler factors gives

$$\begin{aligned}\log|G_\chi(u,t)|&=u\log|L(1+t,\chi)|+\Re R_\chi(u,t),\\ R_\chi(u,t)&=\sum_p\sum_{j\ge2}\frac{(u^j-u)\chi(p)^j}{jp^{j(1+t)}}.\end{aligned}\tag{3.2}$$

Here logarithms of absolute values are real logarithms; the equality follows by taking real parts of the absolutely convergent logarithmic Euler series. Since $0\le u-u^j\le(j-1)(1-u)$,

$$|R_\chi(u,t)|\le(1-u)\sum_{n\ge2}\sum_{j\ge2}n^{-j}=(1-u)\sum_{n\ge2}\frac{1}{n(n-1)}=1-u.\tag{3.3}$$

This is uniform in the character and $t>0$.

For a nonprincipal character, $L(s,\chi)$ is continuous on $[1,2]$ and hence bounded there. Equations (3.2)–(3.3) therefore bound $|G_\chi(u,t)|$ uniformly for $0<u<1$ and $0<t\le1$. No non-vanishing assertion at $s=1$ is used. For the principal character χ_0, put $\kappa_L=\varphi(L)/L$, with $\varphi(1)=1$. The identity

$$L(1+t,\chi_0)=\zeta(1+t)\prod_{p\mid L}(1-p^{-1-t})$$

and $t\zeta(1+t)\to1$ show that $tL(1+t,\chi_0)\to\kappa_L$. Thus, for some fixed $\varepsilon\in(0,1]$ and for $1/2\le u<1$,

$$G_{\chi_0}(u,t)\ge\frac12\left(\frac{\kappa_L}{2t}\right)^u \qquad (0<t\le\varepsilon).\tag{3.4}$$

Indeed, $L(1+t,\chi_0)\ge\kappa_L/(2t)$ there, and $e^{\Re R}\ge e^{-1/2}>1/2$. These standard properties of Dirichlet L-functions follow from their Euler products and the bounded partial sums of nonprincipal characters; see [5, Sections 2 and 6].

Let G_1 denote the unrestricted product (modulus one). For some fixed $\varepsilon_0\in(0,1]$ and $0<t\le\varepsilon_0$, (3.2) gives $G_1(u,t)\le e\zeta(1+t)^u\le 2e\,t^{-u}$. For

every character, the Dirichlet series also gives

$$|G_\chi(u,t)-1| \le \sum_{n\ge 2} n^{-1-t} \ll 2^{-t} \qquad (t\ge 1),$$

uniformly in u. Continuity handles the intervening compact interval. Thus $G_1(u,t)-1$ is integrable on $(0,\infty)$ for each fixed $u<1$. Tonelli's theorem for its nonnegative series and $|\chi(n)|\le 1$ justify termwise integration:

$$J_\chi(u) := \int_0^\infty (G_\chi(u,t)-1)\,dt = \sum_{n\ge 2} \frac{u^{\Omega(n)}\chi(n)}{n\log n}, \tag{3.5}$$

with absolute convergence.

3.2. **The principal character and the residue classes.** If b is invertible modulo L, character orthogonality gives

$$A_{L,b}(u) = \frac{1}{\varphi(L)} \sum_{\chi \bmod L} \overline{\chi(b)} J_\chi(u).$$

For nonprincipal χ, the preceding estimates make $J_\chi(u)$ bounded uniformly in u. For χ_0, the Dirichlet series shows that the integrand is nonnegative, so integrating (3.4) over $(0,\varepsilon)$ gives

$$J_{\chi_0}(u) \ge \frac{(\kappa_L/2)^u \varepsilon^{1-u}}{2(1-u)} - \varepsilon.$$

Consequently

$$(1-u)J_\chi(u) \to 0 \quad (\chi\ne\chi_0), \qquad \liminf_{u\uparrow 1}(1-u)J_{\chi_0}(u) \ge \frac{\kappa_L}{4}. \tag{3.6}$$

In the orthogonality identity, the principal contribution therefore has normalized lower limit at least $\kappa_L/(4\varphi(L)) = 1/(4L)$, while every other normalized term tends to zero. Taking real parts proves (3.1) for coprime residues, including $L=1$.

It remains to treat residues that are not coprime to the modulus. Writing $n=dm$ reduces the congruence to a coprime one; the resulting change from $\log m$ to $\log(dm)$ has bounded total cost. More precisely, choose $0\le b<L$ and put $d=\gcd(b,L)$, $M=L/d$ and $c=b/d$. The residue c is invertible modulo M, including the case $M=1$. For $m\ge 2$,

$$0 \le \frac{1}{m\log m} - \frac{1}{m\log(dm)} \le \frac{\log d}{m(\log m)^2}.$$

The series $C_2=\sum_{m\ge 2} 1/(m(\log m)^2)$ converges by the integral test. Using $\Omega(dm)=\Omega(d)+\Omega(m)$ and $u^{\Omega(m)}\le 1$ therefore gives

$$A_{L,b}(u) \ge \frac{u^{\Omega(d)}}{d} A_{M,c}(u) - \frac{\log d}{d} C_2.$$

The possible omitted term $m=1$, with $d\ge 2$, is nonnegative. After multiplication by $1-u$, the error tends to zero and the leading coefficient tends

to $1/d$. The coprime case gives a lower limit of at least $1/(4dM) = 1/(4L)$, proving (3.1) for every residue.

3.3. **Extracting a finite primitive set.**

Proof of Lemma 2.1. Fix a period L of E. Summing (3.1) over its allowed residue classes gives

$$\liminf_{u\uparrow 1}(1-u)\sum_{\substack{n\geq 2\\ n\in E}} u^{\Omega(n)}\mathrm{w}(n) \geq \frac{\mathrm{d}(E)}{4} > \frac{1}{6}.$$

For each $r \geq 0$, let f_r be the weight of the corresponding level in E:

$$f_r = \sum_{\substack{n\geq 2,\ n\in E\\ \Omega(n)=r}} \mathrm{w}(n).$$

Every f_r is finite, since the convergent series at $u = 1/2$ majorizes $2^{-r}f_r$. Regrouping the nonnegative series gives

$$\sum_{\substack{n\geq 2\\ n\in E}} u^{\Omega(n)}\mathrm{w}(n) = \sum_{r\geq 0} u^r f_r.$$

If $f_r \leq 1/6$ for all $r \geq R$, then

$$(1-u)\sum_{r\geq 0} u^r f_r \leq \frac{1}{6} + (1-u)\sum_{r<R} f_r,$$

contradicting the preceding lower limit. Hence $f_r > 1/6$ for arbitrarily large r. Choose one with $r \geq N \geq 2$. Since $f_r > 1/6$, there is a finite set $P \subset \{n \in E : \Omega(n) = r\}$ with $W(P) > 1/6$. Every element of P is at least $2^r \geq N$, and the common positive value of Ω makes P primitive. This proves the lemma. □

4. Formalization and reproducibility

The companion project proves Theorem 1.1 in Lean 4.24.0. The following table summarizes the formal statements. The four names in the table belong to the namespace `Erdos143`.

Mathematical content	**Declaration**
Countable rational set in $[2,\infty)$, separation and infinite weight.	`mainConclusion_proved`
Adds local finiteness and the displacement bound (1.4).	`strongConclusion_proved`
Main conclusion in $\mathbb{R}$, retaining rationality and asserting nonsummability.	`realConclusion_proved`
One infinite set as the union of nested finite sets satisfying (1.5).	`manuscript_full_statement`

The supplementary manuscript audit checks the last row. Divergence is expressed by an infinite nonnegative sum and, in the real-domain theorem, by nonsummability.

Two analytic proofs of Lemma 2.1 are included. The original uses individual Euler-factor inequalities. Six new modules formalize Section 3, from the double-series convergence and (3.3) through integrability and the character limits to finite selection. The closing theorem in `LogarithmicProof.lean` proves the same selection statement. A dependency audit checks that it uses the logarithmic remainder and does not invoke the original Euler-factor comparisons.

The construction's parameters and invariants are unchanged. The code's `H`, `G` and `N` correspond to p, q and qDn; its denominator is $pq-1$. Extensions use classical choice; no executable construction or explicit parameter bounds are asserted.

The manuscript and sources are deposited at doi:10.5281/zenodo.23676536. The code repository contains reproduction instructions and the detailed correspondence. The project pins mathlib to revision

`f897ebcf72cd16f89ab4577d0c826cd14afaafc7`,

with transitive revisions in `lake-manifest.json`. From the project root, the commands

```
bash scripts/bootstrap.sh
bash scripts/verify.sh
bash scripts/verify-manuscript.sh
```

retrieve the pinned environment and run the build, axiom audits and expanded statement checks, including the logarithmic dependency audit. The current audit covers all 319 proved declarations; the supplementary manuscript statement is checked separately. Only `propext`, `Classical.choice` and `Quot.sound` occur as axiom dependencies. The records distinguish local checks from GitHub builds. Lean and mathlib are not rebuilt from source; the checks do not replace mathematical assessment of the manuscript.

References

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